

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question
 $(x + 2)(x + 3) = (x - 2)(x - 3) + 10$

Which of the following is a solution to the given equation?

- Answer**
- A. 1
 - B. 0
 - C. -2
 - D. -5

Correct Answer: A

Rationale

Choice A is correct. Applying the distributive property on the left- and right-hand sides of the given equation yields $x^2 + 2x + 3x + 6 = x^2 - 2x - 3x + 6 + 10$, or $x^2 + 5x + 6 = x^2 - 5x + 16$. Subtracting x^2 from and adding $5x$ to both sides of this equation yields $10x + 6 = 16$. Subtracting 6 from both sides of this equation and then dividing both sides by 10 yields $x = 1$.

Choices B, C, and D are incorrect. Substituting 0, -2 , or -5 for x in the given equation will result in a false statement. If $x = 0$, the given equation becomes $6 = 16$; if $x = -2$, the given equation becomes $0 = 30$; and if $x = -5$, the given equation becomes $6 = 66$. Therefore, the values 0, -2 , and -5 aren't solutions to the given equation.